CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

CPU & Control Unit

Lecture No.- 3



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Recap of Previous Lecture



Topic Datapath

Topic Control Unit Organization

Topic Hardwired Control Unit

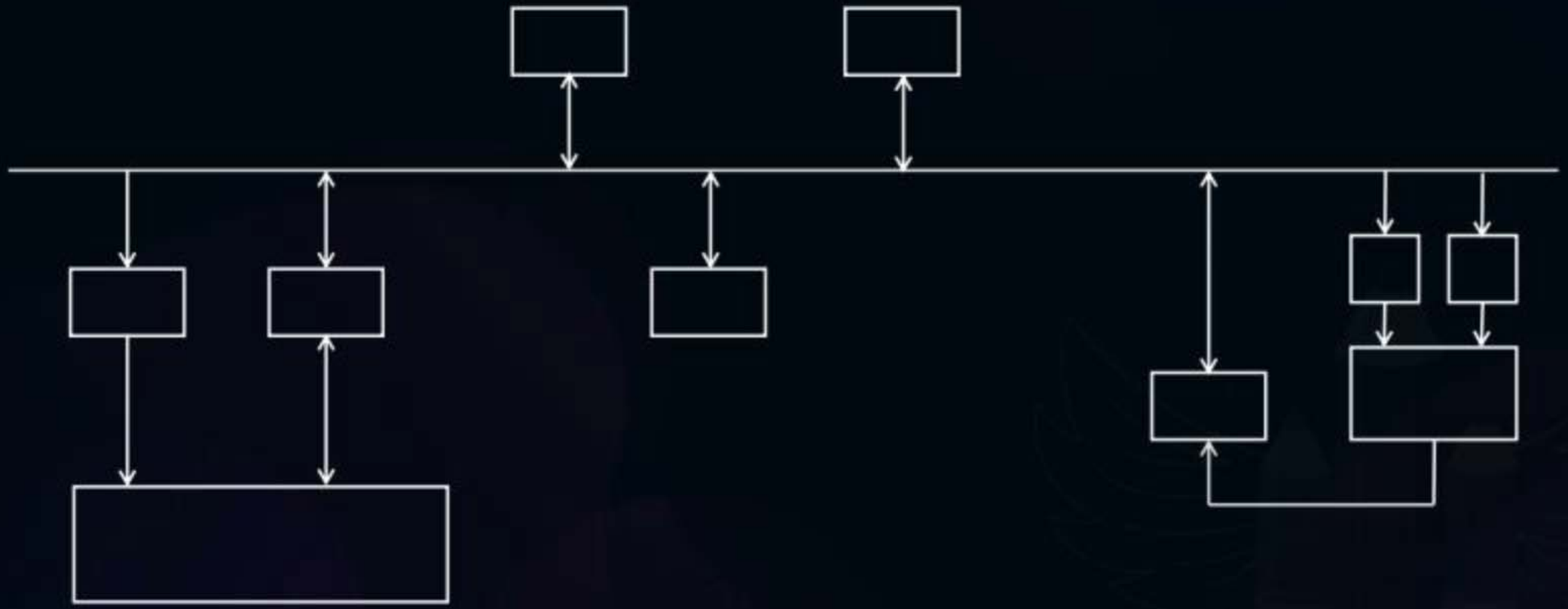
Topics to be Covered



- Topic: Datapath
- Topic: Control Unit Organization
- Topic: Hardwired Control Unit
- Topic: Micro-Programmed Control Unit
- Topic: RISC vs CISC



Topic : Datapath





Topic : Hardwired Control Unit

Control logic is implemented with Gates, flip-flops, decoders and other digital circuits.

Advantage: Can be optimized to produce a faster mode of operation.

Disadvantage: Rearranging the wires among various components is difficult.

Control word

$AR \leftarrow PC$

1 0 0 0 0 - - - 0 | 0 - - - - - - - - 0

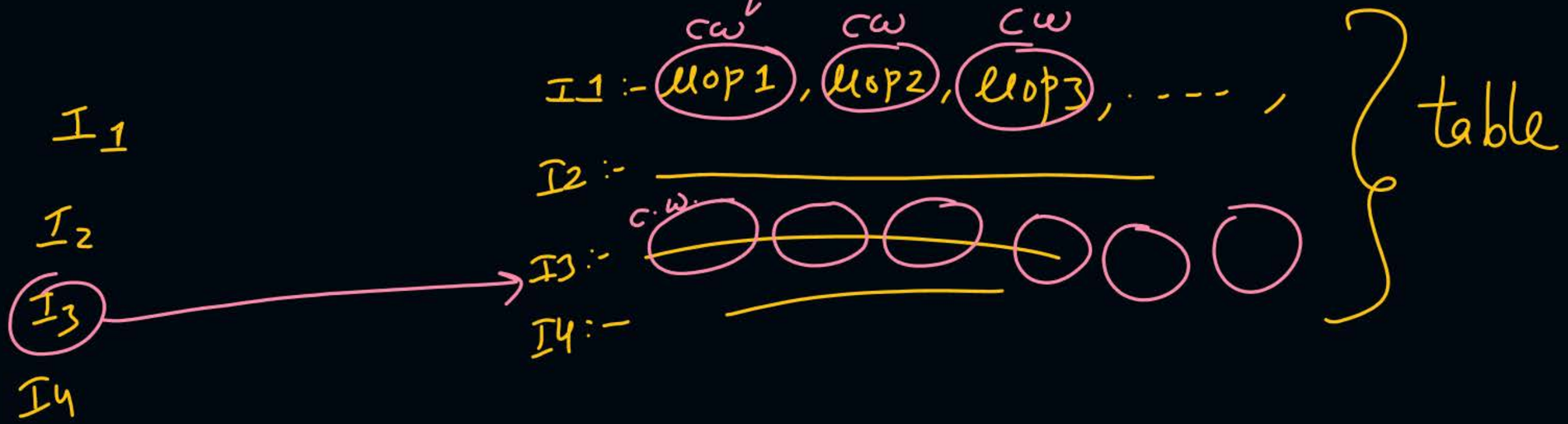
$DR \leftarrow M[AR]$

6 | 0 - - - 0 | 0 - - - - - - - - - - - 0

Control unit designer :-

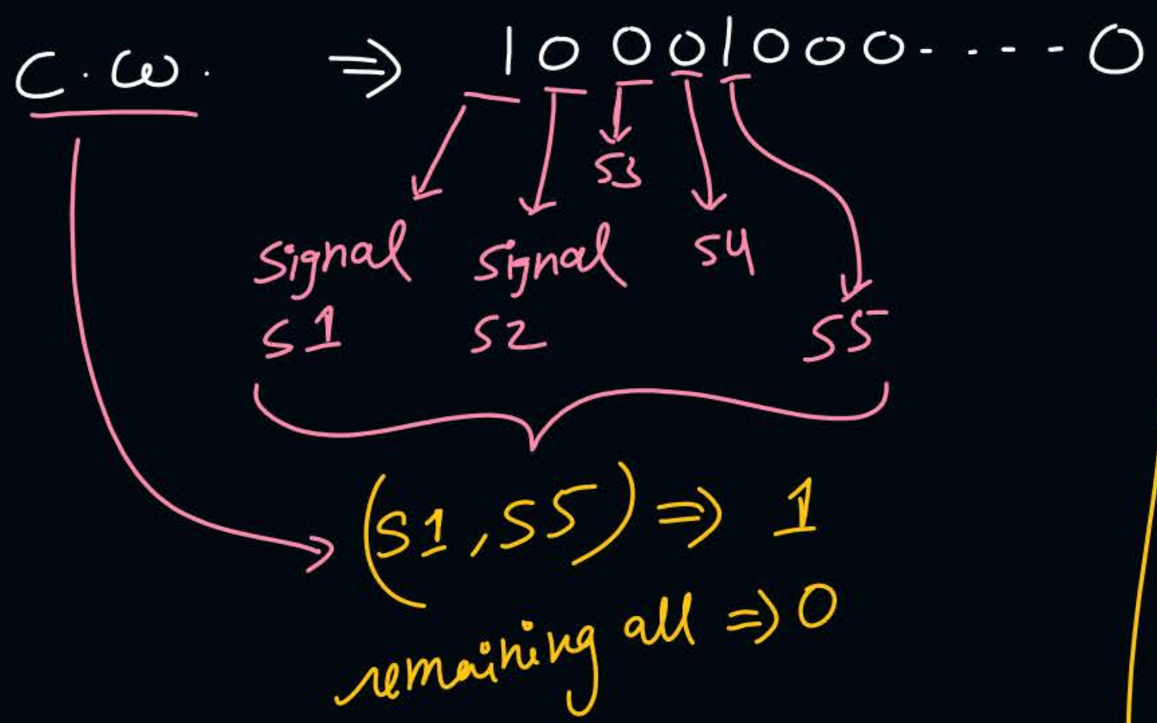
collect all inst^{ns} supported → make a collect^{n} of micro-operat^{ns} sequence for each unique inst^{n}. → for each μ-op^{n} make control word

ex^{s} :-

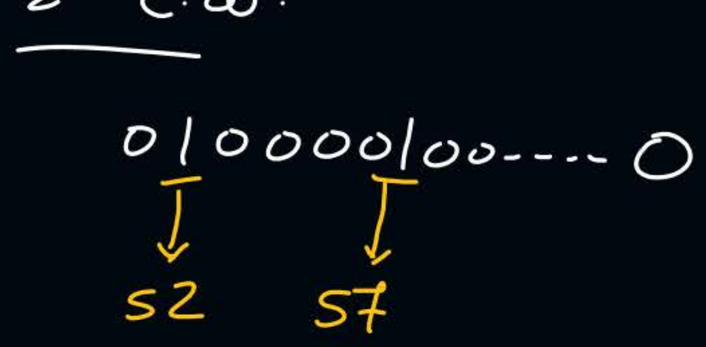


if CPU wants to execute I1,

then first (μ-op) ⇒ ——



2nd C.W.



	Time_1	Time2		Time 8
I1	S1, S5	S2, S7	. . . - - -	S1
I2	. - - - -	ST -	- - -	.
I3	. - - -	- - -	S1 -	.
I4	. - -	- -	- -	S1

Si ⇒ represents control signal

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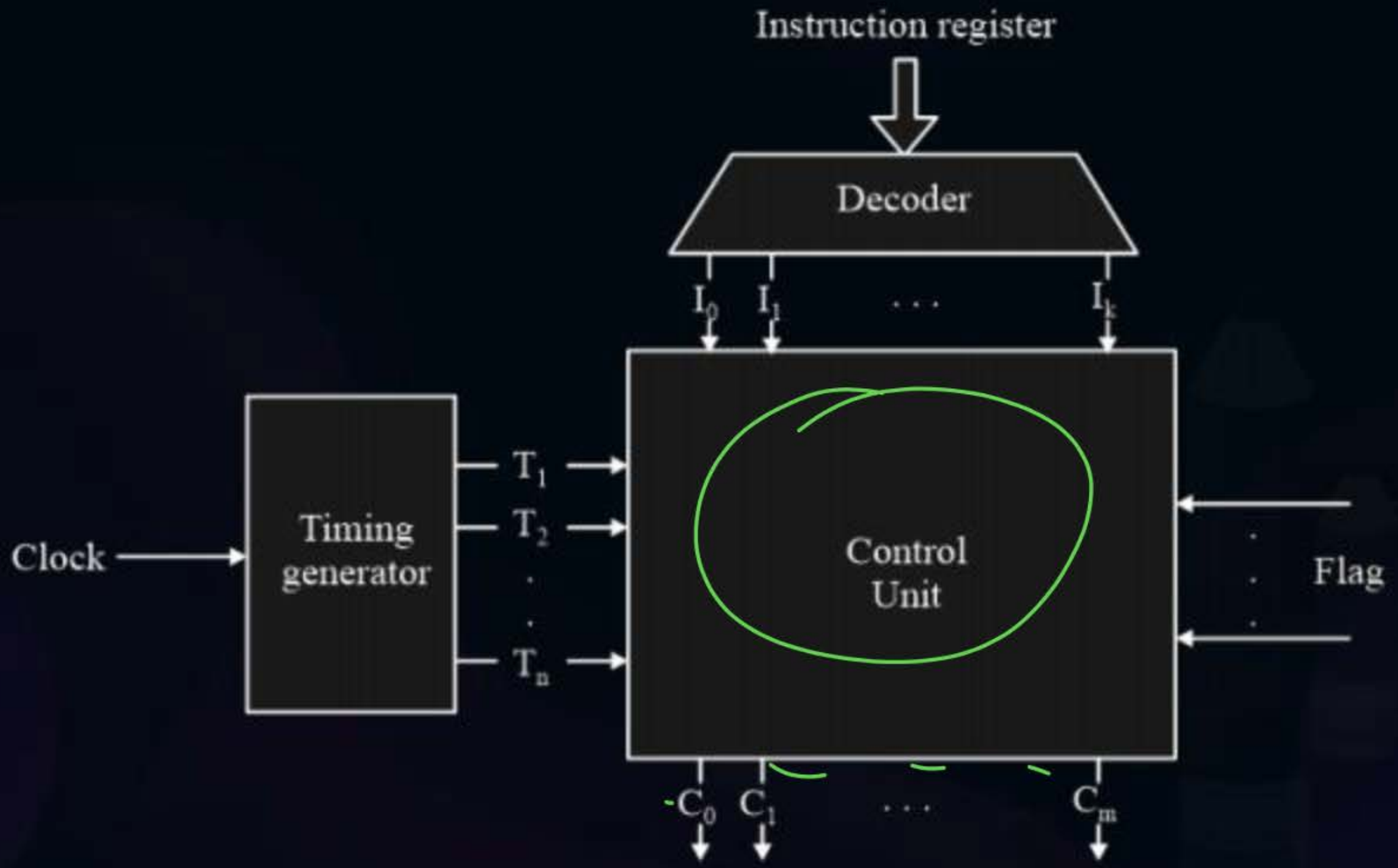
write and minimize boolean expression for each signal.

$$S1 = I2\,T2 + I3T3 + I2\,T8 + I4T8 + I1T1$$

$$= I2T2 + I3T3 + (I2 + I4)\,T8 + I1T1$$



Topic : Hardwired Control Unit



[MCQ]

#Q. A hardwired CPU uses 10 control signals S1 to S10, in various time steps T1 to T5, to implement 4 instructions I1 to I4 as shown below:

	T1	T2	T3	T4	T5
I1	S1, S3, S5	S2, S4, S6	S1, S7	S10	S3, S8
I2	S1, S3, S5	S8, S9, S10	S5, S6, S7	S6	S10
I3	S1, S3, S5	S7, S8, S10	S2, S6, S9	S10	S1, S3
I4	S1, S3, S5	S2, S6, S7	S5, S10	S6, S9	S10

Which of the following pairs of expressions represent the circuit for generating control signals S5 and S10 respectively?

$S_5 = T_1 + T_3(I_2 + I_4)$

A $S5 = T1+I2\cdot T3$ and
$S10 = (I1 + I3)\cdot T4 + (I2 + I4)\cdot T5$

B $S5 = T1+(I2+I4)\cdot T3$ and
$S10 = (I1 + I3)\cdot T4 + (I2 + I4)\cdot T5$

C $S5 = T1+(I2+I4)\cdot T3$ and
$S10 = (I1 + I3 + I4)\cdot T2 + (I2 + I3)\cdot T4+(I2+I4)\cdot T5$

D ✓ $S5 = T1+(I2+I4)\cdot T3$ and
$S10 = (I2 + I3)\cdot T2+I4\cdot T3+(I1 + I3)\cdot T4+(I2+I4)\cdot T5$



Topic : Micro-Programmed Control Unit

all the possible control words are stored in a memory and signals generated as per requirement by reading memory.

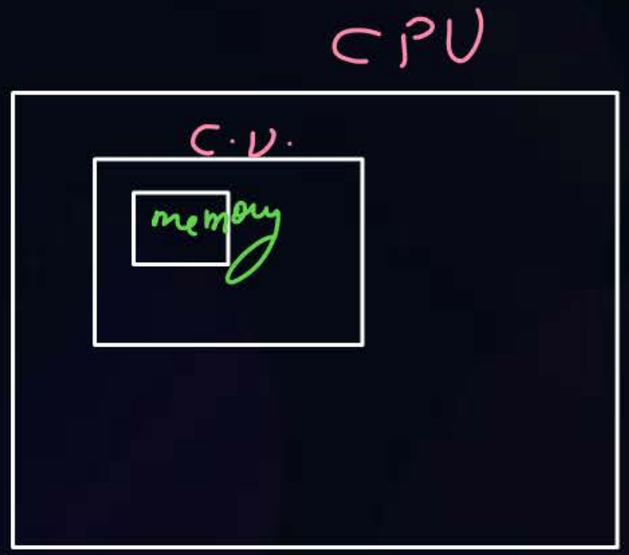
Control logic is implemented with micro-programs.

1.

Advantage: Updating the control logic is easy.

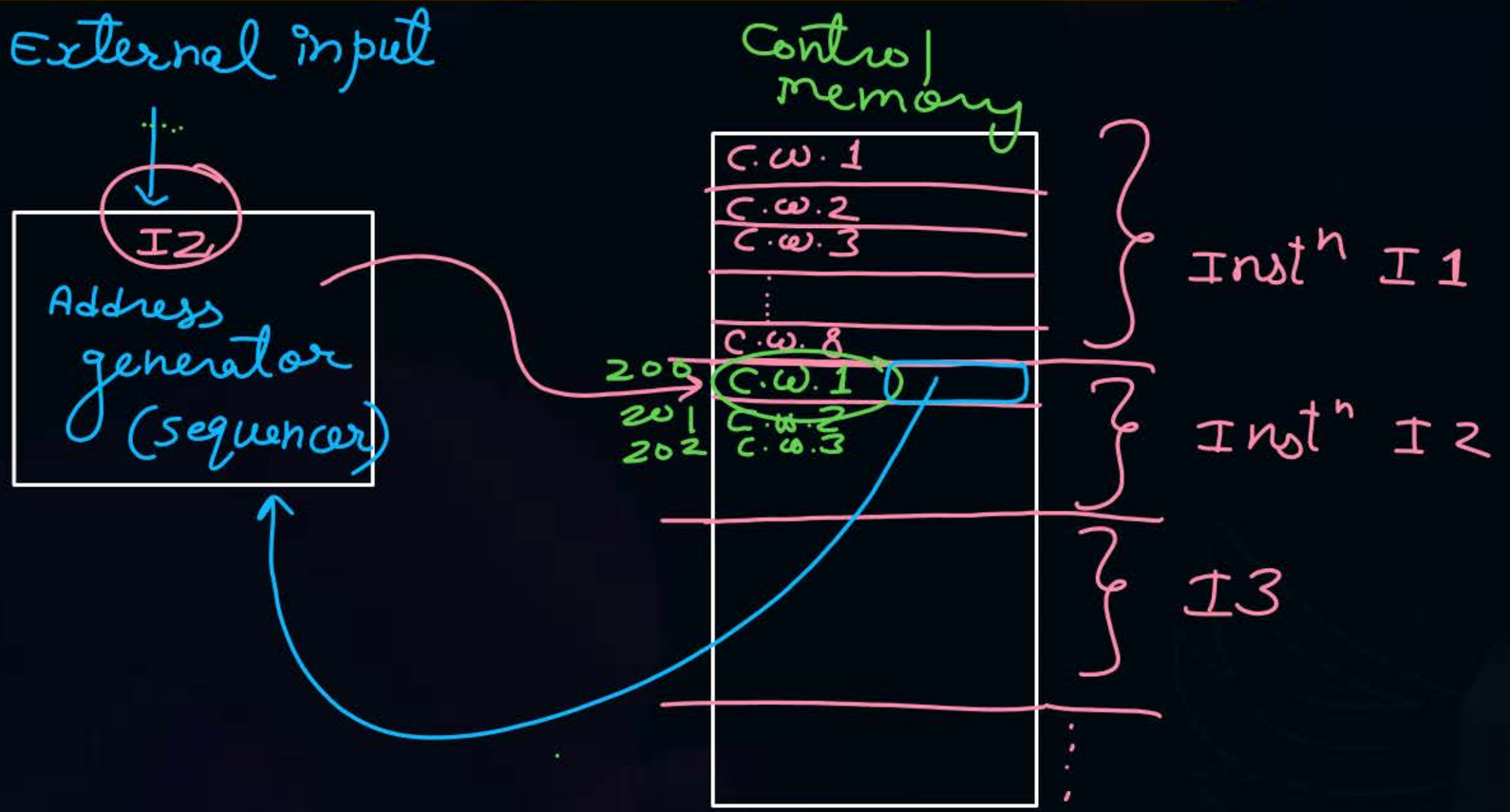
2. can be implemented easily for complex computer.

Disadvantage: Slower than hardwired control unit.



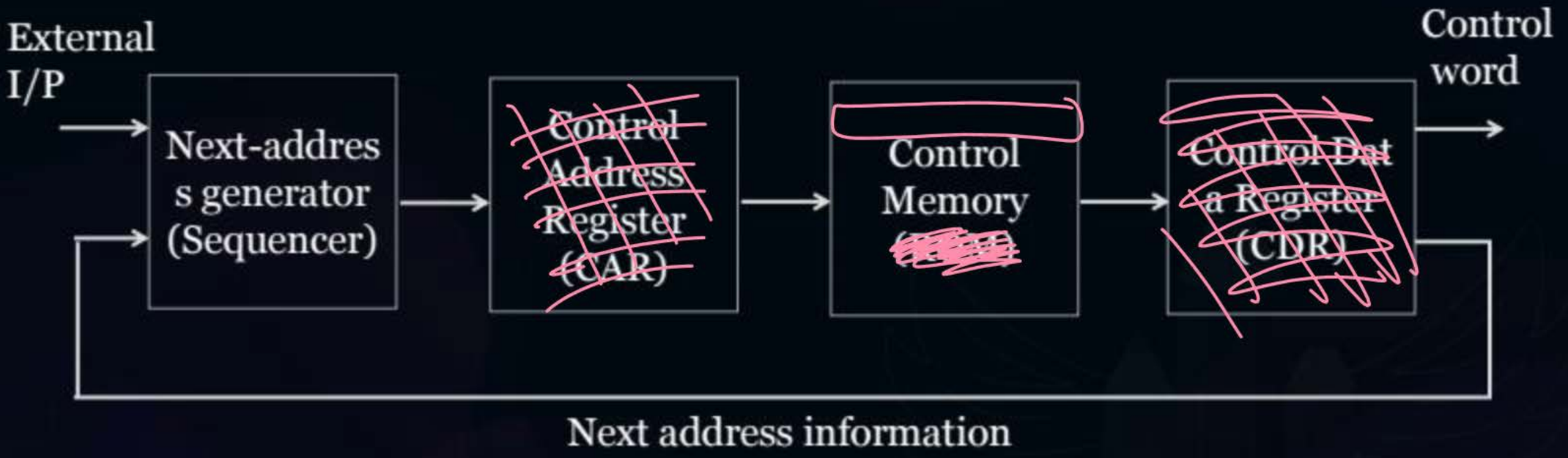


Topic : Micro-Programmed Control Unit

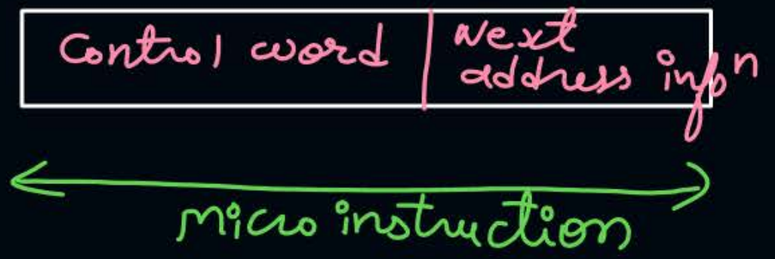




Topic : Control Word Sequencing

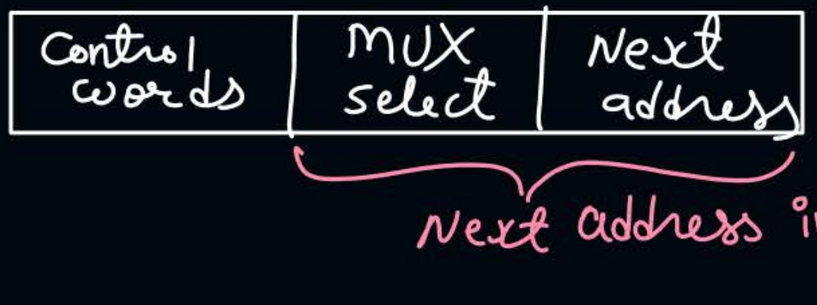


In control mem. at one address



In memory on each address one microinst^n is stored.

Standard format of microinst^n



no. of bits in MUX select
$= \log_2$ (no. of inputs MUX)

No. of mem. addresses in control memory
= no. of microinst^n

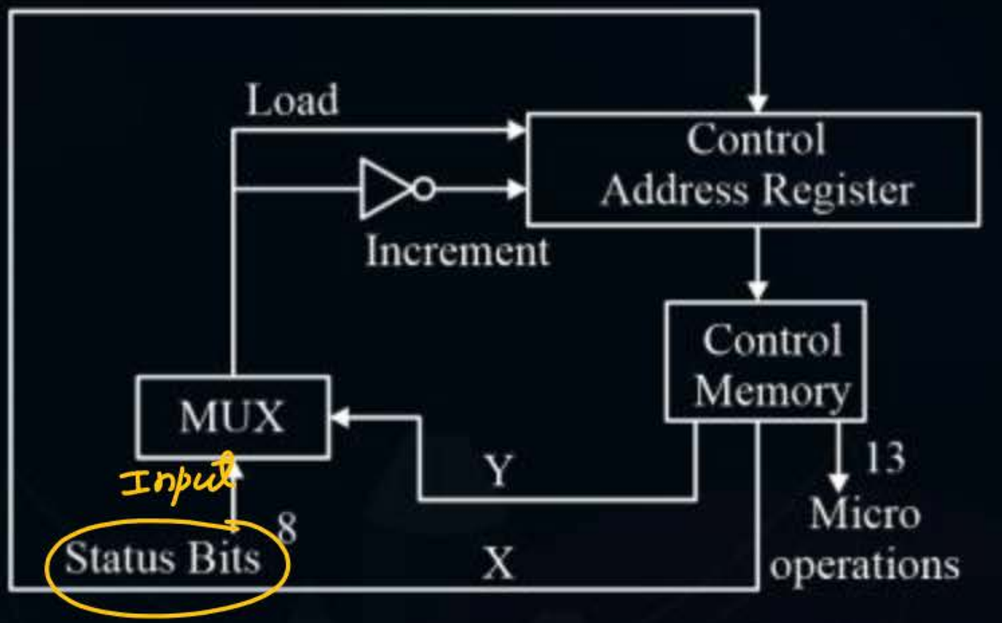
address bits $= \lceil \log_2$ (no. of microinst^n in memory) $\rceil$

[MCQ]

#Q. The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields: a micro-operation field of 13 bits, a next address field (X), and a MUX select field (Y). There are 8 status bits in the inputs of the MUX.

↳ mux select = 3 bits

How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

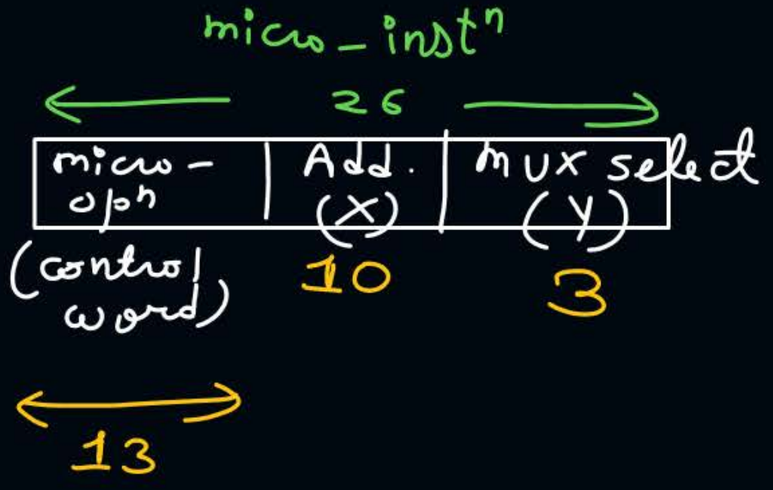


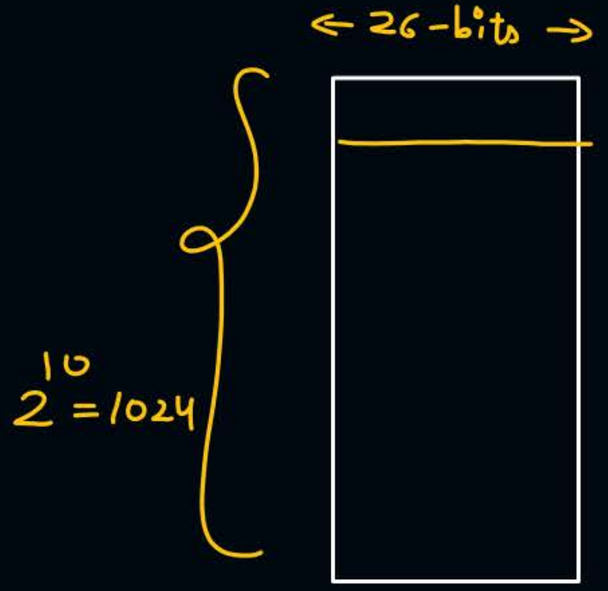
A ✓ 10, 3, 1024

B 8, 5, 256

C 5, 8, 2048

D 10, 3, 512





Control memory size = $(2^{10} * 26 \text{ bits})$

= 26 k bits



Topic : Types of Microprogrammed Control Unit

Horizontal

→ In control word, one bit per signal is stored.

→ Larger size of control word

→ Faster than vertical

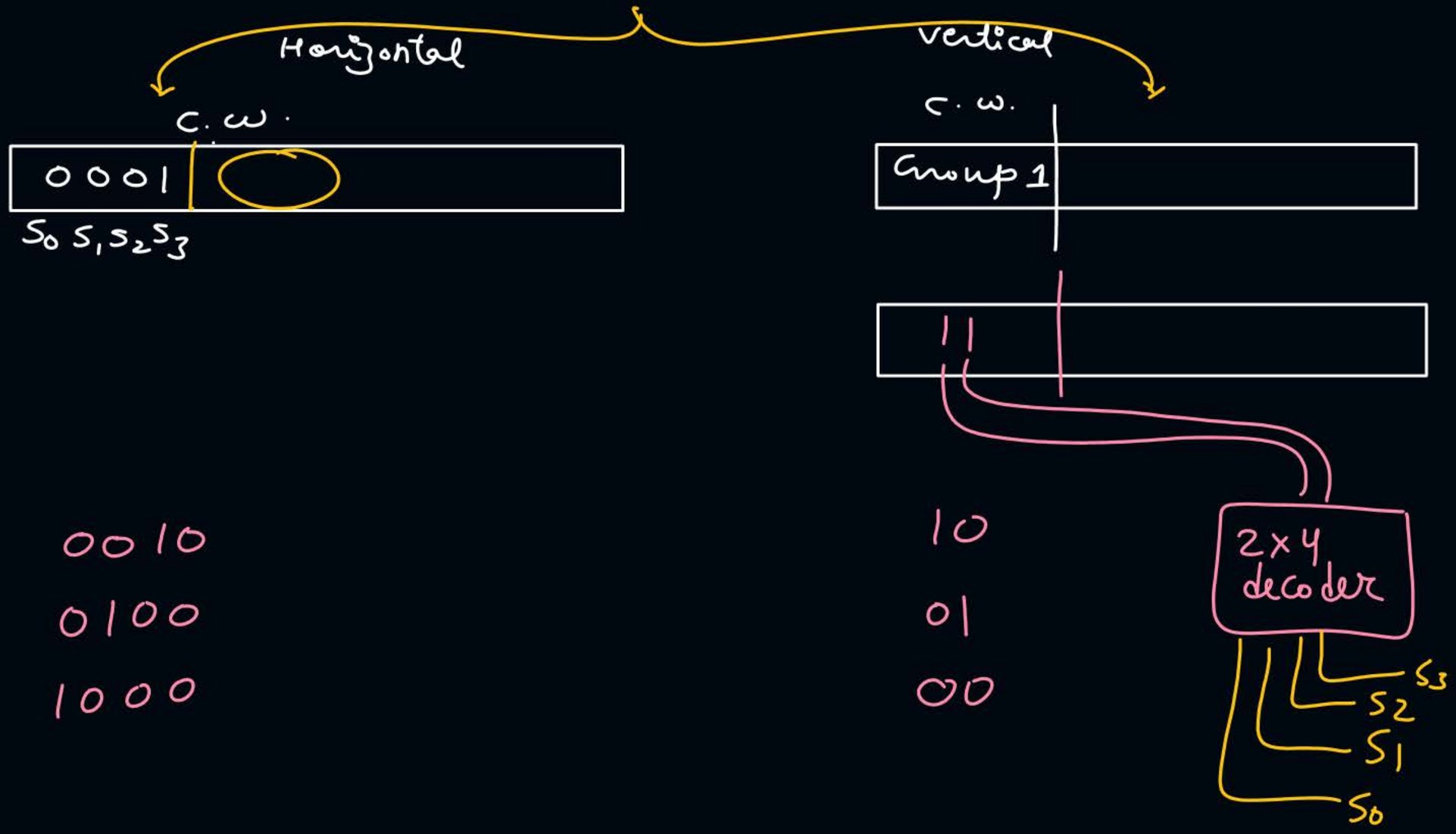
Vertical

→ Signals are divided into multiple groups in such a way that from each group one signal is active at a time. each group infⁿ is stored in encoded form.

→ Smaller size control word

→ To generate signals decoders are used.

→ Slower than horizontal


Horizontal
Vertical
C.W.
0001
S_0 S_1 S_2 S_3
C.W.
Group 1
11
10
01
00
2×4 decoder
S_3
S_2
S_1
S_0
0010
0100
1000



Topic : Speed Comparison

fastest

Hardwired > Horizontal microprogrammed > vertical microprogrammed

slowest

[NAT]

#Q. A control unit generates 120 control signals, which are divided into 6 groups of mutually exclusive signals as below:

Group1 = 30 $\Rightarrow \lceil \log_2 30 \rceil$ = 5 bits

Group2 = 13 $\Rightarrow$ 4 bits

Group3 = 12 $\Rightarrow$ 4 bits

Group4 = 3 $\Rightarrow$ 2 bits

Group5 = 27 $\Rightarrow$ 5 bits

Group6 = 35 $\Rightarrow$ 6 bits

Total = 26 bits

Horizontal = 120 bits

vertical = 26 bits

no. of bits saved = 94

How many bits can be saved by using vertical micro-programmed control unit as compared to horizontal one? 94

[NAT]

H.W.

#Q. A micro-programmed control unit is required to generate a total of 25 control signals. Assume that during any microinstruction at most 2 control signals are active. Minimum number of bits required in the control word to generate the required control signals will be?



2 mins Summary

- Topic: Datapath
- Topic: Control Unit Organization
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- Topic: Micro-Programmed Control Unit
- Topic: RISC vs CISC



Happy Learning

THANK - YOU